



Robotic Emulation of Human Failure Comprehension for Vastly Enhanced Resilience through Metacognition

Fact Sheet

Project Information

RECOVER.ME

Grant agreement ID: 101116620

DOI 

[10.3030/101116620](https://doi.org/10.3030/101116620) 

EC signature date

3 April 2024

Start date

1 July 2024

End date

30 June 2029

Funded under

European Research Council (ERC)

Total cost 

€ 1 499 250,00

EU contribution 

€ 1 499 250,00

Investment in EU policy priorities 

Digital agenda 

Clean air 

Artificial
Intelligence 

Climate action 

Biodiversity 

Coordinated by

DEUTSCHES ZENTRUM FÜR
LUFT - UND RAUMFAHRT EV
Germany

Project description

Initiative for R&I awareness in Serbia

Robotic space exploration is essential, as it enables the study of environments that are too dangerous and distant for human explorers. Unfortunately, these hazardous environments can cause serious hardware malfunctions that are irreparable, often resulting in the decommissioning of the robot. The ERC-funded RECOVER.ME project recognises this inefficiency and aims to develop a novel solution for metacognition-enabled failure handling that emulates human failure comprehension. This solution would allow robots to recover from severe but rectifiable hardware malfunctions without external assistance, leading to improved efficiency in future space exploration and infrastructure projects.

Objective

The aim of the RECOVER.ME project is to achieve human ingenuity in dealing with hardware faults in robotic space exploration. The hypothesis of the project is, that as robots acquire human-like metacognitive awareness and metacognitive regulatory abilities, they will be enabled to recover from severe but rectifiable hardware malfunction all by themselves. This is of particular importance to planetary exploration, as a hardware fault need not be the end of a mission. However, as of today, once a hardware malfunction occurs, the remote robot is typically taken out of operation and troubleshooting is done manually. In the future, especially, when more complex robots are deployed to construct planetary infrastructure for crewed exploration, this can no longer be tolerated. Considering that a hardware fault may occur at any time, such a situation can become safety-critical for the robot, the established infrastructure, and for astronauts in the vicinity of the robot.

To overcome this issue, RECOVER.ME proposes a novel approach for metacognition-enabled failure handling. Instead of relying on hard-coded recovery strategies by specifying how a robot has to react to a certain sub-system fault, the project aims to bootstrap failure handling as a property of the cognitive architecture of the robot itself. Metacognitive awareness is created through a novel knowledge representation that describes how hardware faults may impact robot capabilities. Metacognitive planning will yield contingency configurations employing abstract, affordance-based first order-logic planning for self-programming. To empower robots to monitor their own programming and evaluate the best strategy to react to arbitrary failure cases, generic limitation models

will translate sub-symbolic fault information into semantically interpretable knowledge for metacognitive monitoring and metacognitive evaluation. This will provide robots with competent strategies to deal with faults in a similar way to humans.

Fields of science (EuroSciVoc) ⓘ

This project has not yet been classified with EuroSciVoc.
Be the first one to suggest relevant scientific fields and help us improve our classification service



Keywords ⓘ

Metacognition

Failure Handling

Space Exploration

Programme(s) ⓘ

HORIZON.1.1 - European Research Council (ERC)

MAIN PROGRAMME

[See all projects funded under this programme](#)

Topic(s) ⓘ

[ERC-2023-STG - ERC STARTING GRANTS](#)

[See all projects funded under this topic](#)

Funding Scheme

HORIZON-ERC - HORIZON ERC Grants

[See all projects funded under this funding scheme](#)

Call for proposal

[ERC-2023-STG !\[\]\(c694a3ff3b077d76910920a6a1593ab4_img.jpg\)](#)

[See all projects funded under this call](#)

Host institution



DEUTSCHES ZENTRUM FUR LUFT - UND RAUMFAHRT EV

Net EU contribution 

€ 1 499 250,00

Total cost 

€ 1 499 250,00

Address

LINDER HOHE

51147 KOLN

Germany 

Region

Nordrhein-Westfalen > Köln > Köln, Kreisfreie Stadt

Activity type

Research Organisations

Links

[Contact the organisation !\[\]\(8b0a097b4b9c9c3eeaea0f4289ea77e5_img.jpg\)](#) [Website !\[\]\(87b60107e7fb2186baec8b76a60234e5_img.jpg\)](#)

[Participation in EU R&I programmes !\[\]\(a2bb1e57b467f1e41142026aa73db90f_img.jpg\)](#)

[HORIZON collaboration network](#) 

Beneficiaries (1)



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Germany

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€ 1 499 250,00

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